

Teaching Tech Together

Assessment



Greg Wilson
<https://third-bit.com>



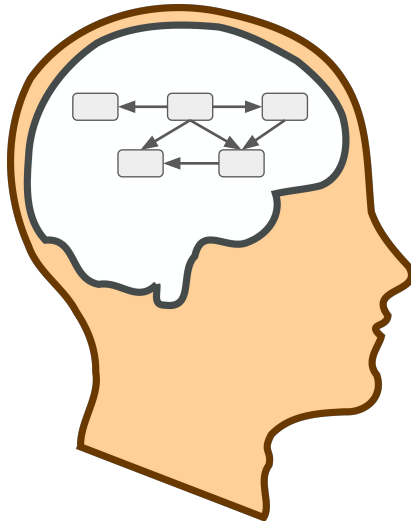
Estimated time: 40 minutes.

All art from openclipart.org (open license)

Learning Objectives

1. Define the terms “formative assessment” and “summative assessment” and explain the different roles of each.
2. Explain why and how formative assessments should probe for misconceptions and describe three ways that instructors can determine what misconceptions to probe for.
3. Explain the complementary purposes of formative assessments for instructors and learners.
4. Explain what peer instruction is, why it is effective, and the requirements for it to be successful.
5. Describe the role of summative assessment and the characteristics of effective summative assessments.
6. Describe the process of reverse instructional design.

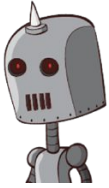
How Can You Tell?



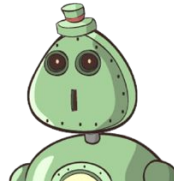
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Once you have a concept map that shows you what you want to teach, you'll be tempted to start writing lessons. Resist: you're not ready yet. Instead, what you need to do next is figure out how you will tell if learning is taking. You need a feedback loop that will tell you when you can move on, or when you have to double back and re-explain something because your learners haven't understood it or have misunderstood.

What Is $37 + 15$?



a) 52



c) 412



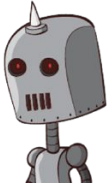
b) 42



d) 43

To see how to do this, let's imagine we're teaching some children how to do multi-digit addition, and we ask them to add 37 and 15.

What Is $37 + 15$?



a) 52

Correct

If they say “52”, then either they understand how to do it or they’ve made a lucky guess.

What Is $37 + 15$?



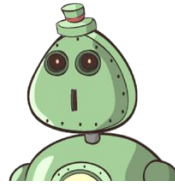
b) 42

Didn't carry

If they got “42”, they probably didn’t carry. In fact, they probably wrote down “12” as the sum of 7 and 5, and then added 3 and 1 to get 4 and wrote that on top of the “1” from “12”.

What Is $37 + 15$?

Added
columns
separately



c) 412

What if they got 412? In this case, they still don't understand how to carry, but it's a step forward: they're keeping the 1.

What Is $37 + 15$?

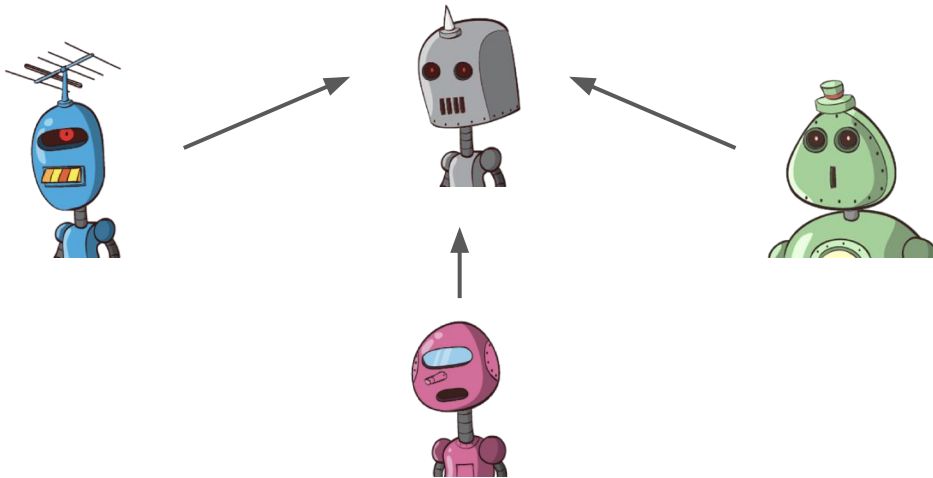
Carried into
the same
column



d) 43

And what about 43? This child has *almost* got it: they've carried the 1, but they've carried it back into the 1's column because they don't understand it's a 10.

Diagnosing Misconceptions



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Each of the learners who gave a wrong answer has a different misconception. Each one therefore needs a different explanation to move forward. It might be a linear progression or it might not; what's important is that if you don't understand what they don't understand, you're just guessing about what to say next.

Description: friendly gray robot (representing the correct answer) in the upper middle. The other three friendly robots (representing wrong answers) are arranged around it. Arrows connect the robots representing wrong answers to the robot representing the correct answer.

Formative Assessment

*Forms (shapes) learning
as it is taking place*

Using questions or exercises during the lesson in order to figure out if learners are forming the right mental models are called *formative assessment*.

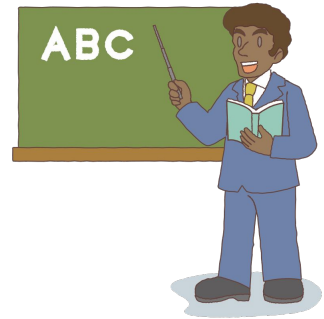
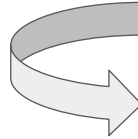
Formative Assessment

- Quick (1-2 minutes)
- Unambiguous (scalable)
- Diagnostic power

Good formative assessments are quick, so that they don't derail teaching. They are unambiguous, so that they'll scale, and they have *diagnostic power*, which means that they give you insight into why things have gone off track.

Formative Assessment

- Quick (1-2 minutes)
- Unambiguous (scalable)
- Diagnostic power

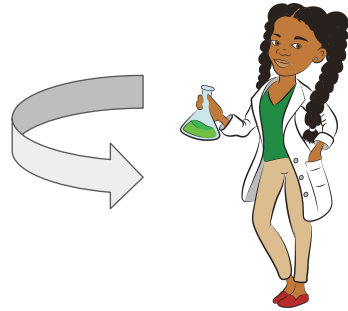


“Can I move on?”

They tell instructors whether they can move on or not (and if not, why not).

Formative Assessment

- Quick (1-2 minutes)
- Unambiguous (scalable)
- Diagnostic power



“Do I understand this?”

And they give learners a way to check that they actually understand something (because it's easy to believe you do until you try to use knowledge).

Peer Instruction



<https://youtu.be/2LbuoxAy56o>

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Taking formative assessment to its logical conclusion leads to *peer instruction*, which is the most effective scalable teaching technique we know.

1. Give a brief introduction to the topic, either in class or in out-of-class reading.
2. Give learners a multiple-choice question (MCQ).
3. Have all the students vote on their answers to the MCQ.
 - a. If all the students have the right answer, move on.
 - b. If they all have the same wrong answer, address that specific misconception.
 - c. If they have a mix of right and wrong answers, give them several minutes to discuss those answers with one another in small groups (typically 2–4 students) and then reconvene and vote again.

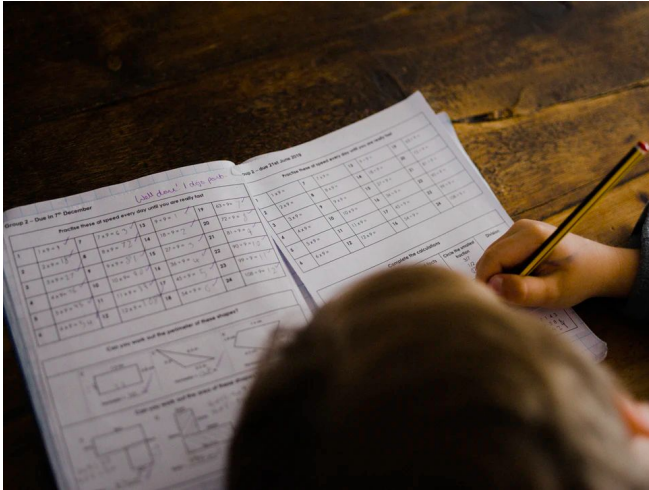
We will look later at how to use this in a programming class.

Exercise

Create a multiple choice question to check something you could teach in two minutes. Include one right answer and two wrong ones, and explain what misconceptions the wrong answers are meant to diagnose.

Create a multiple choice question to check understanding of one of these topics. Include one right answer and two wrong ones, and explain what misconceptions the wrong answers are meant to diagnose.

Alternative exercises



- MCQ
- Parsons Problem
- Fill in the blanks
- ***Label diagrams***

Having students label diagrams can give a lot of insights too, but we don't have good online tools for it (yet).

Image <https://unsplash.com/photos/JexAuNCfefs>

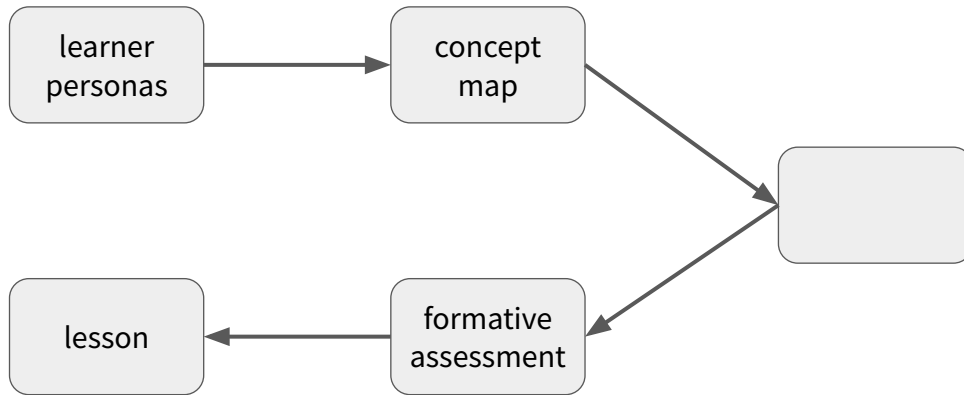
Exercise

How can you figure out what misconceptions your learners are likely to have when you're building a lesson for the very first time and haven't met them?



How can you figure out what misconceptions your learners are likely to have?

Reverse Instructional Design



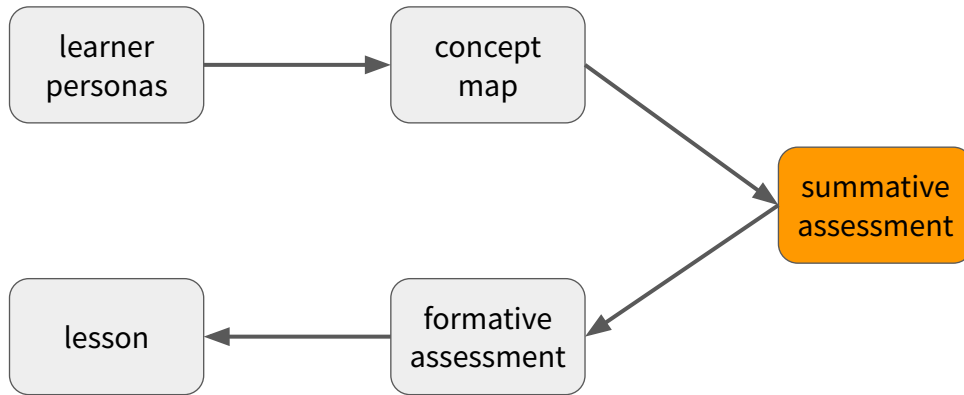
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A lot of people write lessons and then figure out how they're going to check for learning. It turns out be more effective to go in reverse: figure out how you're going to check that people have understood the topics you're planning to teach, then put those formative assessments in some sort of order and work backward to connect them with lessons. This style of lesson development was invented independently at least three times, and has several names—we like to call it *reverse instructional design*. There's still one component we need to add to complete the picture, though.

Description: a five-stage workflow. Each stage is a box connected to the next by a single arrow. The five stages are labeled:

1. Learner personas
2. Concept map
3. Blank (unlabeled)
4. Formative assessment
5. Lesson

Summative Assessment

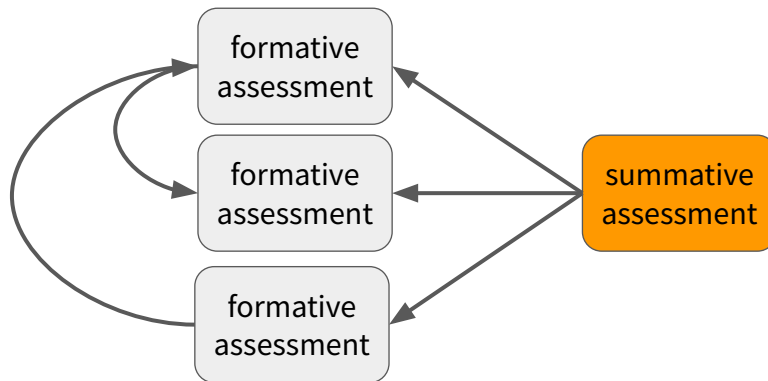


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That missing component is called *summative assessment*. It's like a driving exam—its purpose is to tell you if the learner has mastered the material and is ready to go out on their own.

Description: the same five-stage workflow as the previous slide. The third step, which was previously an unlabeled box, is now labeled “summative assessment” and highlighted.

Summative Assessment



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One reason to create the summative assessment at this stage is that it helps you figure out what formative assessments you need. Your learners should have a chance to practice everything they're going to do in the summative assessment *before* they do it. Breaking that final assessment down into component skills tells you what formative assessments you need; figuring out their dependencies tells you the order in which you should use them.

Description: a single highlighted box on the right labeled "summative assessment" is connected by arrows to a stack of three boxes, each of which is labeled "formative assessment". Curved arrows connect some of the formative assessment boxes to each other to show an order in which they might be placed.

Lesson Design: PETE

1. Problem
2. Explanation
3. Theory
4. Example

Here's one pattern for designing an explanation.

Lesson Design: PRIMM

1. Predict
2. Run
3. Investigate
4. Modify
5. Make

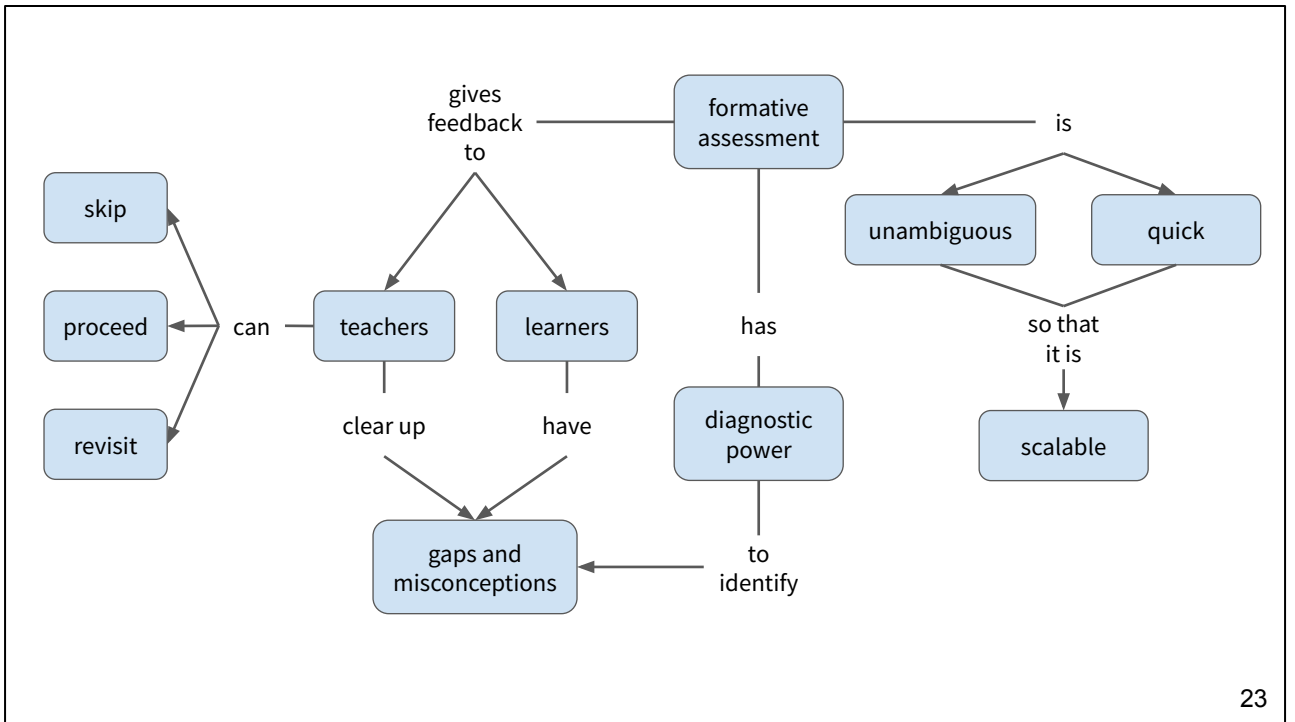
Here's another pattern.

Discussion

What do you do if 80% of the class answers a formative assessment correctly but 20% doesn't? What about a 90/10 split or 99/1? Discuss in small groups and then describe your strategies to the class.



Formative assessments tell you how well things are going, but you still need to decide what to do with that knowledge. In particular, when do you circle back and when do you move on? If only one person doesn't understand something, do you want the rest of the class to sit there and get bored while you explain? Instructors make decisions like these all the time; in groups, discuss what your thresholds and strategies are, and then share your ideas with the class.



Concept map for formative assessment.

Description: concept map.